

Problem Statement

Project Title

Customer Churn Prediction Using Machine Learning

Problem Statement

Customer churn is one of the biggest challenges faced by subscription-based businesses such as telecommunications, banking, insurance, and online service providers. Acquiring new customers is significantly more expensive than retaining existing ones. Therefore, organizations need an effective way to identify customers who are likely to discontinue their services.

The objective of this project is to develop a machine learning model that predicts whether a customer is likely to churn based on demographic information, account details, service usage, and billing history. By analyzing customer behavior and historical data, the model aims to identify high-risk customers before they leave the company.

The project involves data preprocessing, exploratory data analysis (EDA), feature engineering, handling class imbalance using SMOTE, and training multiple machine learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost. These models are evaluated using performance metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score to determine the most effective model for churn prediction.

The final solution enables businesses to proactively implement customer retention strategies, improve customer satisfaction, reduce revenue loss, and support data-driven decision-making.

Objectives

The primary objectives of this project are:

- To analyze customer demographic, service, and account-related data.
 - To identify the factors that influence customer churn.
 - To perform data cleaning, preprocessing, and feature engineering.
 - To handle class imbalance using the SMOTE technique.
 - To build and compare multiple machine learning classification models.
 - To evaluate model performance using standard classification metrics.
 - To select the best-performing model for customer churn prediction.
 - To provide actionable business insights for improving customer retention.
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Business Problem

Many customers discontinue telecom services due to factors such as high monthly charges, contract type, lack of value-added services, payment preferences, or poor customer experience. Since companies often fail to identify these customers before they leave, they experience significant revenue loss and increased customer acquisition costs.

This project addresses this challenge by building a predictive model that identifies customers with a high probability of churning. The predictions can help businesses focus their retention efforts on the right customers and improve overall profitability.

Scope of the Project

The project covers the following activities:

- Data collection and understanding
 - Data cleaning and preprocessing
 - Exploratory Data Analysis (EDA)
 - Feature engineering and encoding
 - Handling imbalanced data using SMOTE
 - Feature scaling
 - Model building using:
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - XGBoost
 - Model evaluation and comparison
 - Business insights and recommendations
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Expected Outcome

The expected outcome of this project is a machine learning model capable of accurately predicting customer churn. The model should help businesses identify customers at risk of leaving, allowing them to implement targeted retention strategies and reduce customer attrition.

Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

- Imbalanced-learn (SMOTE)
 - XGBoost
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Dataset Description

The project uses the Telecom Customer Churn dataset, which contains customer demographic information, subscribed services, account details, billing information, and the target variable indicating whether a customer has churned.

Key features include:

- Gender
 - Senior Citizen
 - Partner
 - Dependents
 - Phone Service
 - Multiple Lines
 - Internet Service
 - Online Security
 - Online Backup
 - Device Protection
 - Tech Support
 - Streaming TV
 - Streaming Movies
 - Contract Type
 - Paperless Billing
 - Payment Method
 - Tenure
 - Monthly Charges
 - Total Charges
 - Churn (Target Variable)
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Conclusion

Predicting customer churn using machine learning enables organizations to identify high-risk customers before they discontinue services. By applying predictive analytics, companies can design targeted retention campaigns, reduce customer attrition, improve customer satisfaction, and increase long-term profitability. This project demonstrates how data-driven decision-making can support effective customer relationship management and business growth.